

ENGR 3323: Signals and Systems

HW 8_Ch6 Answer Keys

Q1) For the periodic signals $x(t)$ and $y(t)$ shown below:

a) Here, $T_0 = 10\pi$, $\omega_0 = \frac{2\pi}{T_0} = \frac{1}{5}$. Because of even symmetry, all the sine terms are zero.

$$x(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n}{5}t\right) + b_n \sin\left(\frac{n}{5}t\right)$$

$$a_0 = \frac{1}{5} \quad (\text{by inspection})$$

$$a_n = \frac{2}{10\pi} \int_{-\pi}^{\pi} \cos\left(\frac{n}{5}t\right) dt = \frac{1}{5\pi} \left(\frac{5}{n}\right) \sin\left(\frac{n}{5}t\right) \Big|_{-\pi}^{\pi} = \frac{2}{\pi n} \sin\left(\frac{n\pi}{5}\right)$$

$$b_n = \frac{2}{10\pi} \int_{-\pi}^{\pi} \sin\left(\frac{n}{5}t\right) dt = 0 \quad (\text{integrand is an odd function of } t)$$

Since $b_n = 0$, $C_n = |a_n|$ and $\theta_n = \angle a_n$, both shown in Fig. S6.1-1b. The corresponding frequency ω is easily computed as $\omega_0 n = \frac{n}{5}$.

In this case, $T_0 = 2\pi$, $\omega_0 = 1$. Thus,

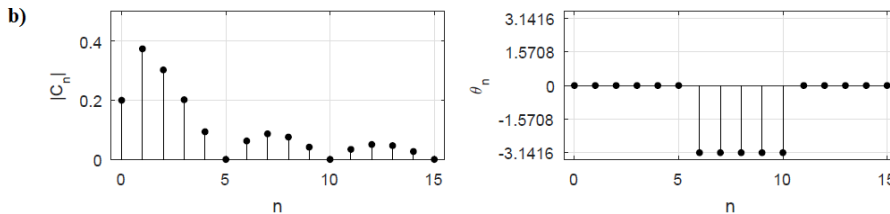
$$y(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos nt + b_n \sin nt \quad \text{with } a_0 = 0.5 \quad (\text{by inspection}),$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \cos nt dt = 0, \quad b_n = \frac{1}{\pi} \int_0^{2\pi} \frac{t}{2\pi} \sin nt dt = -\frac{1}{\pi n}$$

and

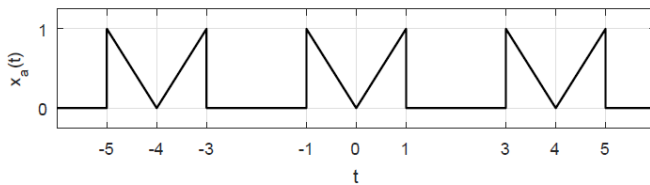
$$y(t) = 0.5 - \frac{1}{\pi} \left(\sin t + \frac{1}{2} \sin 2t + \frac{1}{3} \sin 3t + \frac{1}{4} \sin 4t + \dots \right)$$

$$= 0.5 + \frac{1}{\pi} \left[\cos\left(t + \frac{\pi}{2}\right) + \frac{1}{2} \cos\left(2t + \frac{\pi}{2}\right) + \frac{1}{3} \cos\left(3t + \frac{\pi}{2}\right) + \dots \right]$$

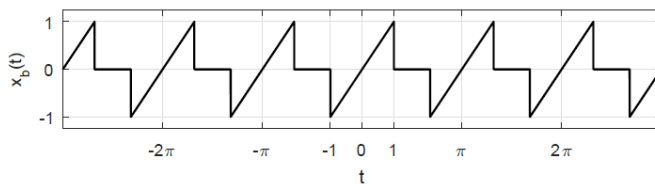


Q2)

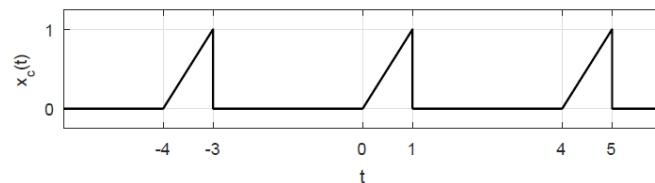
a)



b)



c)



Q3)

In each case the signal is periodic, it is because $x(t) = x(t+T)$ for all t . If $x(t) \neq x(t+T)$ for all t , then the signal is not periodic.

Periodic?	(a)	(b)	(c)	(d)	(e)	(f)
Yes	Yes	Yes	No	Yes	No	Yes
ω_0	1	1		π		1
Period T	2π	2π		2		2π